United States Patent	12409329
Kind Code	B2
Date of Patent	September 09, 2025
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Method and apparatus for implanting a medical device in a coronary sinus

Abstract

A system for delivery of a leadless pacemaker. The system includes a catheter with an elongate flexible tubular body with a proximal end and a distal end, wherein the distal end of the tubular body includes a delivery cup with an external surface having an inflatable compliant balloon; and a pacing capsule of the leadless pacemaker releasably retained in the delivery cup. The pacing capsule includes an arrangement of tines configured to deploy and pierce a target tissue at a desired pacing capsule implant site in a coronary sinus of a patient. The balloon, when at least partially inflated, is configured to urge the delivery cup against the target tissue during deployment of the pacing capsule.

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Appl. No.: **17/813184**

Filed: **July 18, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230098146 A1	Mar. 30, 2023

Related U.S. Application Data

us-provisional-application US 63250431 20210930

Publication Classification

Int. Cl.: **A61N1/375** (20060101); **A61N1/372** (20060101)

U.S. Cl.:

CPC **A61N1/37518** (20170801); **A61N1/37205** (20130101); **A61N1/37512** (20170801); **A61N1/37516** (20170801); **A61N1/3756** (20130101);

Field of Classification Search

CPC: A61N (1/37518); A61N (1/37205); A61N (1/37512); A61N (1/37516); A61N (1/3756)

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/250,431, filed Sep. 30, 2021, the entire content of which is incorporated herein by reference.

BACKGROUND

(1) The cardiac conduction system includes the sinus atrial (SA) node, the atrioventricular (AV) node, the bundle of His, bundle branches and Purkinje fibers. A heartbeat is initiated in the SA node, which may be described as the natural “pacemaker” of the heart. An electrical impulse arising from the SA node causes the atrial myocardium to contract. The electrical impulse, or electrical pulse or signal, is conducted to the ventricles via the AV node which inherently delays the conduction to allow the atria to stop contracting before the ventricles begin contracting thereby providing proper AV synchrony. The electrical impulse is conducted from the AV node to the ventricular myocardium via the bundle of His, bundle branches, and Purkinje fibers.

(2) The leadless pacemaker, which is significantly smaller than a conventional pacemaker coupled to one or more transvenous leads, is a self-contained generator and electrode system configured to be implanted directly into the heart. The leadless pacemaker, which does not utilize leads extend from out of the heart of a patient, eliminates several complications associated with transvenous pacemakers and leads such as, for example, pocket infections, hematoma, lead dislodgment, and lead fracture. The leadless pacemaker also has cosmetic appeal because there is no chest incision or visible pacemaker pocket.

(3) The leadless pacemaker includes one or more electrodes on its outer housing to deliver therapeutic electrical signals and/or sense intrinsic depolarizations of the heart. Intracardiac medical devices may provide cardiac therapy functionality, such as sensing and pacing, and may also be used to treat either atrial or ventricular arrhythmias or fibrillation.

(4) The leadless pacemaker device may be implanted via a femoral vein transcatheter approach and requires no chest incision or subcutaneous generator pocket. The catheter system utilized to deploy the leadless pacemaker includes a distal end with a delivery cup housing the self-contained generator and electrode system, referred to herein as a pacing capsule. The delivery cup is maneuvered into the proper position, e.g., in the right atrium at or near the triangle of Koch (TOK) or the right ventricle, under fluoroscopic visualization or using a sonogram produced by an ultrasound imaging system.

SUMMARY

(5) The distal tip of the delivery cup can be difficult to discern in a sonogram or other image used to monitor the position of the delivery cup during an implantation procedure, and in some cases the pacing capsule itself can cast a shadow that partially or even fully obscures the distal tip of the delivery cup in the sonogram image. In some cases, the difficulties in navigating the pacing capsule to the target tissue location in the TOK result in inaccurate placement of the distal end of the catheter. This inaccurate placement can cause the delivery cup on the distal end of the catheter to slide away from the target tissue in the TOK and enter the right ventricle, which can result in an insecure implantation of the pacing capsule, or even an unsuccessful implantation procedure.

(6) Implantation of a leadless pacemaker device in the coronary sinus may provide an alternative to implantation in the TOK region for delivery of pacing to the left ventricle, the His Bundle, or other parts of the conduction system of the heart. In general, the present disclosure is directed to a system including a catheter for use in the delivery and implantation of a pacing capsule of an implantable medical device in the coronary sinus of a heart of a patient. A distal end of the catheter includes a delivery cup that releasably retains the pacing capsule. The delivery cup includes an inflatable

balloon that, when inflated, urges the delivery cup against a target tissue location in the coronary sinus of the heart. While the delivery cup is temporarily fixed in position adjacent to the target tissue location by the inflated balloon, the pacing capsule of the leadless pacemaker can be deployed from the delivery cup. When deployed, an arrangement of tines on the pacing capsule pierce and engage the tissue of the coronary sinus to accurately implant the pacing capsule into an implant location in the coronary sinus of the patient.

(7) In one example embodiment, the system of the present disclosure allows a practitioner to navigate the distal end of the catheter containing the delivery cup and the pacing capsule to a desired implant location in the coronary sinus with the balloon in an uninflated or partially inflated state. The balloon may then be inflated to apply a force against a first wall of the coronary sinus and temporarily fix the delivery cup into a deployment position against a second wall opposite the first wall. After the pacing capsule is deployed from the delivery cup and pierces the second wall to anchor the pacing capsule into the target tissue, the balloon may be deflated to facilitate removal of the catheter from the vasculature of the patient.

(8) In many patients, in a sonogram image the coronary sinus is a more easily identifiable structure within the heart than the TOK, and cannulating the coronary sinus using a sonogram can be more reliable and reproducible for practitioners than locating the TOK. Once the desired implant location has been identified and the distal end of the catheter moved into position, the inflated balloon urges the delivery cup into a fixed deployment position in the coronary sinus, which can avoid dislocation or slippage of the delivery cup away from the TOK and into the right atrium or right ventricle. Temporarily anchoring the delivery cup in the coronary sinus with the inflated balloon makes implantation of the pacing capsule more reliable and reproducible, and places the implanted pacing capsule in a more effective position closer to the left ventricular tissue in the heart of the patient.

(9) In one aspect, the present disclosure is directed to a system for delivery of a leadless pacemaker. The system includes a catheter with an elongate flexible tubular body with a proximal end and a distal end, wherein the distal end of the tubular body includes a delivery cup with an external surface having an inflatable compliant balloon; and a pacing capsule of the leadless pacemaker releasably retained in the delivery cup, wherein the pacing capsule includes an arrangement of tines configured to deploy and pierce a target tissue at a desired pacing capsule implant site in a coronary sinus of a patient. The balloon, when at least partially inflated, is configured to urge the delivery cup against the target tissue during deployment of the pacing capsule.

(10) In another aspect, the present disclosure is directed to a method for implanting a leadless pacemaker in a target cardiac tissue. The method includes inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system including: a catheter having an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup includes an inflatable compliant balloon; and a pacing capsule releasably retained in the delivery cup, wherein the pacing capsule includes an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue, wherein the target cardiac tissue is a wall of a coronary sinus; maneuvering the delivery cup in the vasculature of the patient to an implant location in the coronary sinus of the patient; at least partially inflating the balloon to urge the delivery cup against the target cardiac tissue and anchor the delivery cup at the implant location; and deploying the pacing capsule from the delivery cup such that the tines pierce the target tissue to embed the electrode in the target tissue and affix the pacing capsule in the target tissue at the implant location.

(11) In another aspect, the present disclosure is directed to a method for implanting a leadless pacemaker in a target cardiac tissue in a coronary sinus of a patient. The method includes: inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system including: a catheter having an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup includes an inflatable balloon; and a

pacing capsule releasably retained in the delivery cup, wherein the pacing capsule includes an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue; maneuvering the delivery cup in the vasculature of the patient to an implant location in a first wall of the coronary sinus of the patient; inflating the balloon to urge the delivery cup against a second wall of the coronary sinus opposite a first wall and temporarily affix the delivery cup at the implant location; and deploying the pacing capsule from the delivery cup such that the tines pierce and embed in the first wall to implant the pacing capsule in the first wall, and the electrode embeds in the first wall.

(12) The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic illustration of a system for guiding and implanting a leadless pacemaker into a target tissue.

(2) FIGS. **2**A-**2**D are conceptual diagrams illustrating a pacing capsule for a leadless pacemaker.

(3) FIG. **3** is a conceptual circuit diagram of circuitry utilized in a housing of the pacing capsule of an embodiment of a leadless pacemaker of FIGS. **2**A-**2**D.

(4) FIG. **4** is a schematic illustration of the pacing capsule of FIGS. **2**A-**2**D affixed to a target tissue at a desired implant location in a coronary sinus of a heart of a patient.

(5) FIG. **5** is a conceptual diagram of a map of a heart of a patient showing implantation locations in the coronary sinus for use with, e.g., the illustrative systems and devices of the present disclosure.

(6) FIG. **6** is a conceptual diagram of a map of a heart of a patient showing implantation locations in the coronary sinus for use with, e.g., the illustrative systems and devices of the present disclosure.

(7) FIG. **7** is a cross-sectional view of a leadless pacemaker implantation system including a delivery cup with a compliant balloon and configured to releasably retain the pacing capsule of FIGS. **2**A-**2**D.

(8) FIGS. **8**A-**8**B are schematic illustrations of a distal end of a pacemaker implantation system.

(9) FIG. **9** is a schematic illustration of the delivery cup of the system of FIG. **7** during an implantation process at a target tissue in a desired implant location in a coronary sinus of a heart of a patient.

(10) FIG. **10** is a flow chart of an illustrative example of a method for implanting a leadless pacemaker utilizing the devices of FIGS. **7-9**.

(11) Like symbols in the drawings indicate like elements.

DETAILED DESCRIPTION

(12) FIG. **1** is a schematic illustration (which is not to scale) of a system **30** for guiding a pacing capsule of a leadless pacemaker to a desired implant location and implanting the pacing capsule into a target cardiac tissue. The system **30** includes an elongate tubular catheter **32** having a body **34** with a bore **39** that extends from a proximal end **31** to a distal end **33** thereof. In some examples, which are not intended to be limiting, the catheter body **34** has a length of about 75 centimeters (cm) to about 150 cm.

(13) The proximal end **31** of the catheter body **34** is connected to a control handle **36** that can be used to deflect the catheter body **34** and deploy a pacing capsule **40**. The pacing capsule **40** is releasably retained at the distal end **33** of the catheter body **34** in a delivery cup **42**. Suitable leadless pacemaker pacing capsules **40** include, but are not limited to, those available from

Medtronic, Inc., Minneapolis, MN, under the trade designation MICRA. The pacing capsule **40** includes a self-contained generator and electrode system that is implantable into cardiac tissue, and do not require leads or a subcutaneous pacemaker pocket like a transvenous pacemaker system.
(14) The catheter body **34** may be placed in a suitable vein of a patient such as, for example, the femoral vein, and moved through the venous system to place a distal end **48** of the delivery cup **42** at a predetermined location in the heart. In some examples, the catheter body **34** may be deflected using an optional curve deflection control **50** on the handle **36**. In some examples, the location of the catheter body **34** and a distal region **48** of the delivery cup **42** is monitored with an ultrasonic imaging system, and a sonogram image of the catheter body **34** and the delivery cup **42** is used to precisely position the delivery cup **42** in the heart. In some embodiments, during placement procedures a proximal portion of the pacing capsule **40** can optionally be tethered via a mechanical tether (not shown in FIG. **1**) such as, for example, a tether pin **47**.
(15) Once the delivery cup **42** is positioned within the coronary sinus of the heart, the pacing capsule **40** is deployed from the delivery cup **42** via manipulation of the pacing capsule **40** or engagement of an optional deployment control **52** on the handle **36**. The pacing capsule **40** can be implanted into the cardiac tissue using, for example, an arrangement of spring-loaded or self-expanding metal tines, a screw-in metal helix, and combinations thereof. After the pacing capsule **40** is implanted in the target tissue of the heart, a tether lock **54** on the handle **36** is released, and the catheter body **34** is withdrawn from the vascular system of the patient. The pacing capsule remains in position within the heart to provide suitable pacing therapy for the patient.
(16) In some examples, the handle **36** may optionally include a fluid port **56**, which can provide a fluid flush through the bore **39** of the catheter body **34** using a fluid such as, for example, water, saline, and the like.
(17) In some examples (not shown in FIG. **1**), the catheter **32** may optionally be placed within a rigid introducer including a tapered distal dilating tip to ease introduction of the catheter body **34** into the vasculature of the patient. For example, the dilating tip may be formed from an elastomeric material such as a silicone. In some examples, the introducer may optionally be coated with a lubricious hydrophilic coating.
(18) FIG. **2**A is a conceptual diagram illustrating a plan view of an example of a relatively compact pacing capsule **140** including a fixation component **144**. The pacing capsule **140** of FIG. **2**A includes a housing **146** extending along a longitudinal axis LA from a proximal end **151** to a distal end **153**. The housing **146** may be formed from a biocompatible and biostable metal such as titanium, and in some examples, the housing **146** may be hermetically sealed. The housing **146** may include any suitable dimensions, and in some examples, has an outer diameter of between about 10 French (Fr) and about 30 Fr, such as about 20 Fr.
(19) The pacing capsule **140** may contain electronic circuitry, including one or more of sensing circuitry (e.g., for sensing cardiac signals), therapy delivery circuitry (e.g., for generating cardiac pacing pulses), and processing circuitry for controlling the functionality of the pacing capsule **140**, and may include one or more leadlets **148**. The leadlet **148** may terminate in a distal electrode **149**. For example, the leadlet **148** may include a conductor, such as an electrically conductive material, extending through a non-conductive jacket, such as polytetrafluoroethylene (PTFE) coating or polyether ether ketone (PEEK) tube, a portion of the conductor being exposed at electrode **149**. The leadlet **148** may be used to control unipolar or bipolar pacing and sensing of different selected tissues. In some examples, the leadlet **148** may include an optional insulative layer, for example, medical grade polyurethane, parylene, or a silicone.
(20) In some examples, the pacing capsule **140** can optionally include a second electrode (meant for surface atrial pacing, not shown in FIGS. **2**A-**2**D) placed between the fixation tines **164**. The second electrode can provide dual chamber pacing while the pacing capsule **140** lays against the tissue in the coronary sinus of the patient.
(21) The electronic circuitry in the pacing capsule **140** may be configured to generate and deliver

an electrical pulse therapy to tissue proximate leadlet **148** via first electrode **149**, through the tissue to a return electrode **155**. The leadlet **148** may be spaced apart from a distal end **153** of the housing **146**, for example, being coupled to the sensing and therapy delivery circuitry by the conductor of a hermetic feedthrough assembly (not shown).

(22) In some examples, the pacing capsule **140** includes an optional retrieval structure **158** fixedly attached to the proximal end **151** of the housing **146**. The retrieval structure **158** may be configured for temporarily tethering the pacing capsule **140** to a delivery catheter or a retrieval catheter, or may be configured to couple to mechanical tether assemblies.

(23) The fixation component **144** includes an arrangement of tines **160**. In the embodiment of FIG. **2**A, the fixation component **144** includes a base **159** affixed to the housing **146** of the pacing capsule **140**, a penetrator tine **162**, one or more optional fixation tines **164**, and an optional support tines **170**. However, as shown in more detail below, the fixation component **144** may include a wide variety of tine shapes, numbers of tines, and the like. Some examples of suitable tine arrangements, shapes and the like may be found in U.S. application Ser. No. 16/990,239, which is incorporated by reference herein in its entirety. The tines **160** may include one or more preset curved sections and one or more optional substantially straight sections. For example, the penetrator tine **162** may include preset curvature **166**, and fixation tine **164** may include a preset curvature **168**. In some examples, the tines **160** may define a ribbon shape configured to deform along a plane normal to longitudinal axis LA and resist twisting outside of that plane.

(24) As further illustrated in FIGS. **2**B-**2**D, the penetrating tine **162** is configured to guide the pacing capsule **140** to the target tissue area, while the fixation tine **164** supports the penetrating tine **162** as the electrode **149** penetrates the target tissue. The fixation tines **164** are biased to one side of the body **146** of the pacing capsule **140**, which allows the pacing capsule **140** to implant when the pacing capsule **140** is parallel to the tissue (rather than requiring a substantially perpendicular approach). In some example embodiments, the fixation tines **164** can be anywhere from 5-90 degrees apart and may not be required if the penetrating tine **162** can properly prevent dislodgment of the pacing capsule **140** from a target tissue area. In some examples, the fixation tines **164** allow the pacing capsule **140** to implant substantially parallel to the target tissue. In various examples, to facilitate parallel implantation, the pacing capsule **140** can include no fixation tines **164**, or up to 2 fixation tines **164** on a side of the pacing capsule **140**, as well as an arrangement of penetrating tines **162** and optional support tines **170**.

(25) For example, the fixation tine **164** may be the same or substantially similar to tines described in U.S. Published Application No. 2020/0306522, which is incorporated by reference herein in its entirety. The fixation tines **164** may be configured to have a target deflection stiffness and a target deployment stiffness, which may include a measure of a resistance to force applied to the pacing capsule **140** in a proximal direction when fixation component **144** is engaged with tissue at a target implantation site. In some examples, the target deflection stiffness may be selected to enable the fixation tines **164** to deflect a predetermined amount to enable visualization of the fixation tines **164** under fluoroscopy. In some examples, the target deflection stiffness of the fixation tines **164** may be about 0.2 N to about 0.8 N, such as about 0.3 N to about 0.6 N. The deployment stiffness may include a measure of a force applied by the fixation tines **164** as the fixation tines **164** move from a deformed configuration to an undeformed configuration when the fixation component **144** is deployed from a distal opening of the delivery cup **42** (FIG. **1**) such that the free distal ends **163**, **165** of the tines **160** penetrates cardiac tissue. In some examples, the target deployment stiffness may be within a range from about 0.6 N to about 1.2 N.

(26) The tines **160** may include any suitable elastically deformable biocompatible material. In some examples, the tines **160** may include a super-elastic material, such as, for example, a nickel titanium alloy. For example, the fixation component **144** may be cut from a medical grade nickel titanium alloy tubing that conforms to the chemical, physical, mechanical, and metallurgical requirements of the ASTM F2063 standard, and has a wall thickness of about 0.005 inch (0.127

mm). In this way, the tines **160** may be integrally formed with the base **159**, and each tine of tines **160** may have a thickness of about 0.005 inch±0.001 inch (0.127 mm±0.0254 mm). In some examples, after cutting the tubing or otherwise forming fixation component **144**, the tines **160** may be shaped into a preset configuration by bending and holding tines **160**, while heat treating the construction according to methods known to those skilled in the art.
(27) FIGS. **2**B-**2**D illustrate different views of the pacing capsule **140**. In some examples, as illustrated in FIG. **2**C, the penetrator tine **162** may define an aperture **180**. The aperture **180** may be configured to control a deployment stiffness or deflection stiffness of penetrator tine **162**. For example, the aperture **180** may define a shear stress reduction region configured to reduce shear stress in penetrator tine **162** when bent, such as when bent into the deformed configuration or during a pull test or tug test.
(28) FIG. **3** is a block diagram of circuitry that may be enclosed within the housing **146** (FIG. **2**A) of the pacing capsule **140** to provide the functions of calibrating pacing therapy and/or delivering pacing therapy, using the device **10** according to one example. The electronic circuitry enclosed within the housing **146** may include software, firmware, and hardware that cooperatively monitor atrial and ventricular electrical cardiac signals, determine when a cardiac therapy is necessary, and/or deliver electrical pulses to the patient's heart according to programmed therapy mode and pulse control parameters. The electronic circuitry may include a control circuit **80** (e.g., including processing circuitry), a memory **82**, a therapy delivery circuit **84**, a sensing circuit **86**, and/or a telemetry circuit **88**. In some examples, the device **10** includes one or more sensors **90** for producing a signal that is correlated to a physiological function, state, or condition of the patient, such as a patient activity sensor, for use in determining a need for pacing therapy and/or controlling a pacing rate. For example, one sensor **90** may include an inertial measurement unit (e.g., accelerometer) to measure motion.
(29) The power source **98** may provide power to the circuitry of the device **10** including each of the components **80**, **82**, **84**, **86**, **88**, **90** as needed. The power source **98** may include one or more energy storage devices, such as one or more rechargeable or non-rechargeable batteries. The connections (not shown) between the power source **98** and each of the components **80**, **82**, **84**, **86**, **88**, **90**, may be understood from the general block diagram illustrated to one of ordinary skill in the art. For example, the power source **98** may be coupled to one or more charging circuits included in the therapy delivery circuit **84** for providing the power used to charge holding capacitors included in the therapy delivery circuit **84** that are discharged at appropriate times under the control of the control circuit **80** for delivering pacing pulses, e.g., according to a dual chamber pacing mode such as DDI(R). The power source **98** may also be coupled to components of the sensing circuit **86**, such as sense amplifiers, analog-to-digital converters, switching circuitry, etc., sensors **90**, the telemetry circuit **88**, and the memory **82** to provide power to the various circuits.
(30) The functional blocks shown represent functionality included in the device **10** and may include any discrete and/or integrated electronic circuit components that implement analog, and/or digital circuits capable of producing the functions attributed to the medical device **10** herein. The various components may include processing circuitry, such as an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group), and memory that execute one or more software or firmware programs, a combinational logic circuit, state machine, or other suitable components or combinations of components that provide the described functionality. The particular form of software, hardware, and/or firmware employed to implement the functionality disclosed herein will be determined primarily by the particular system architecture employed in the medical device and by the particular detection and therapy delivery methodologies employed by the medical device.
(31) The memory **82** may include any volatile, non-volatile, magnetic, or electrical non-transitory computer-readable storage media, such as random-access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM),

flash memory, or any other memory device. Furthermore, the memory **82** may include a non-transitory computer-readable media storing instructions that, when executed by one or more processing circuits, cause the control circuit **80** and/or other processing circuitry to calibrate pacing therapy and/or perform a single, dual, or triple-chamber calibrated pacing therapy (e.g., single or multiple chamber pacing), or other cardiac therapy functions (e.g., sensing or delivering therapy), attributed to the device **10**. The non-transitory computer-readable media storing the instructions may include any of the media listed above.

(32) The control circuit **80** may communicate, e.g., via a data bus, with the therapy delivery circuit **84** and the sensing circuit **86** for sensing cardiac electrical signals and controlling delivery of cardiac electrical stimulation therapies in response to sensed cardiac events, e.g., P-waves and R-waves, or the absence thereof. Electrodes such as, for example, a tip electrode **92**, a distal housing-based electrode **94**, and a proximal housing-based electrode **96** may be electrically coupled to the therapy delivery circuit **84** for delivering electrical stimulation pulses to the patient's heart and to the sensing circuit **86** and for sensing cardiac electrical signals.

(33) The sensing circuit **86** may include an atrial (A) sensing channel **87** and a ventricular (V) sensing channel **89**. The distal housing-based electrode **94** and the proximal housing-based electrode **96** may be coupled to the atrial sensing channel **87** for sensing atrial signals, e.g., P-waves attendant to the depolarization of the atrial myocardium. In examples that include two or more selectable distal housing-based electrodes, the sensing circuit **86** may include switching circuitry for selectively coupling one or more of the available distal housing-based electrodes to cardiac event detection circuitry included in the atrial sensing channel **87**. Switching circuitry may include a switch array, switch matrix, multiplexer, or any other type of switching device suitable to selectively couple components of the sensing circuit **86** to selected electrodes. The tip electrode **92** and the proximal housing-based electrode **96** may be coupled to the ventricular sensing channel **89** for sensing ventricular signals, e.g., R-waves attendant to the depolarization of the ventricular myocardium.

(34) Each of the atrial sensing channel **87** and the ventricular sensing channel **89** may include cardiac event detection circuitry for detecting P-waves and R-waves, respectively, from the cardiac electrical signals received by the respective sensing channels. The cardiac event detection circuitry included in each of the channels **87** and **89** may be configured to amplify, filter, digitize, and rectify the cardiac electrical signal received from the selected electrodes to improve the signal quality for detecting cardiac electrical events. The cardiac event detection circuitry within each channel **87** and **89** may include one or more sense amplifiers, filters, rectifiers, threshold detectors, comparators, analog-to-digital converters (ADCs), timers, or other analog or digital components. A cardiac event sensing threshold, e.g., a P-wave sensing threshold and an R-wave sensing threshold, may be automatically adjusted by each respective sensing channel **87** and **89** under the control of the control circuit **80**, e.g., based on timing intervals and sensing threshold values determined by the control circuit **80**, stored in the memory **82**, and/or controlled by hardware, firmware, and/or software of the control circuit **80** and/or the sensing circuit **86**.

(35) Upon detecting a cardiac electrical event based on a sensing threshold crossing, the sensing circuit **86** may produce a sensed event signal that is passed to the control circuit **80**. For example, the atrial sensing channel **87** may produce a P-wave sensed event signal in response to a P-wave sensing threshold crossing. The ventricular sensing channel **89** may produce an R-wave sensed event signal in response to an R-wave sensing threshold crossing. The sensed event signals may be used by the control circuit **80** for setting pacing escape interval timers that control the basic time intervals used for scheduling cardiac pacing pulses. A sensed event signal may trigger or inhibit a pacing pulse depending on the particular programmed pacing mode. For example, a P-wave sensed event signal received from the atrial sensing channel **87** may cause the control circuit **80** to inhibit a scheduled atrial pacing pulse and schedule a ventricular pacing pulse at a programmed atrioventricular (AV) pacing interval. If an R-wave is sensed before the AV pacing interval expires,

the ventricular pacing pulse may be inhibited. If the AV pacing interval expires before the control circuit **80** receives an R-wave sensed event signal from the ventricular sensing channel **89**, the control circuit **80** may use the therapy delivery circuit **84** to deliver the scheduled ventricular pacing pulse synchronized to the sensed P-wave.

(36) The therapy delivery circuit **84** may include atrial pacing circuit **83** and ventricular pacing circuit **85**. Each pacing circuit **83**, **85** may include charging circuitry, one or more charge storage devices such as one or more low voltage holding capacitors, an output capacitor, and/or switching circuitry that controls when the holding capacitor(s) are charged and discharged across the output capacitor to deliver a pacing pulse to the pacing electrode vector coupled to respective pacing circuits **83**, **85**. The tip electrode **92** and the proximal housing-based electrode **96** may be coupled to the ventricular pacing circuit **85** as a bipolar cathode and anode pair for delivering ventricular pacing pulses, e.g., upon expiration of an AV or VV pacing interval set by the control circuit **80** for providing atrial-synchronized ventricular pacing and a basic lower ventricular pacing rate.

(37) The atrial pacing circuit **83** may be coupled to the distal housing-based electrode **94** and the proximal housing-based electrode **96** to deliver atrial pacing pulses. The control circuit **80** may set one or more atrial pacing intervals according to a programmed lower pacing rate or a temporary lower rate set according to a rate-responsive sensor-indicated pacing rate. Atrial pacing circuit may be controlled to deliver an atrial pacing pulse if the atrial pacing interval expires before a P-wave sensed event signal is received from the atrial sensing channel **87**. The control circuit **80** starts an AV pacing interval in response to a delivered atrial pacing pulse to provide synchronized multiple chamber pacing (e.g., dual- or triple-chamber pacing).

(38) Charging of a holding capacitor of the atrial or ventricular pacing circuit **83**, **85** to a programmed pacing voltage amplitude and discharging of the capacitor for a programmed pacing pulse width may be performed by the therapy delivery circuit **84** according to control signals received from the control circuit **80**. For example, a pace timing circuit included in the control circuit **80** may include programmable digital counters set by a microprocessor of the control circuit **80** for controlling the basic pacing time intervals associated with various single-chamber or multiple-chamber pacing (e.g., dual- or triple-chamber pacing) modes. The microprocessor of the control circuit **80** may also set the amplitude, pulse width, polarity, or other characteristics of the cardiac pacing pulses, which may be based on programmed values stored in the memory **82**.

(39) The device **10** may optionally include other sensors **90** for sensing signals from the patient for use in determining a need for and/or controlling electrical stimulation therapies delivered by the therapy delivery circuit **84**. In some examples, a sensor indicative of a need for increased cardiac output may include a patient activity sensor, such as an accelerometer. An increase in the metabolic demand of the patient due to increased activity as indicated by the patient activity sensor may be determined by the control circuit **80** for use in determining a sensor-indicated pacing rate.

(40) Control parameters utilized by the control circuit **80** for sensing cardiac events and controlling pacing therapy delivery may be programmed into the memory **82** via the telemetry circuit **88**, which may also be described as a communication interface. The telemetry circuit **88** includes a transceiver and antenna for communicating with an external device, such as a programmer or home monitor, using radio frequency communication or other communication protocols. The control circuit **80** may use the telemetry circuit **88** to receive downlink telemetry from and send uplink telemetry to the external device. In some cases, the telemetry circuit **88** may be used to transmit and receive communication signals to/from another medical device implanted in the patient.

(41) FIG. **4** is a conceptual diagram illustrating the pacing capsule **140** implanted at a desired implant site **200** in the coronary sinus of a patient. The tines **160** define a deformable preset curvature configured to position the pacing capsule **140** at selected target tissue of the desired implant site **200**, which is between opposed walls **202**A, **202**B of the coronary sinus. When deployed at target implant site **200**, the tines **162**, **164** have a deployment stiffness that enables the tines **162**, **164** to penetrate selected target ventricular tissue at the desired implant site **200** such that

the distal end **149** of the leadlets **148** is in the ventricular tissue of the wall **202**B. For example, the pacing capsule **140** may be secured at the implant site **200** by the tines **162**, **164** penetrating through the wall **202**B of the coronary sinus. The tines **162**, **164** are configured for spring-loaded release, upon deployment out through distal opening of the delivery cup **142** (FIG. **1**) such that the respective free distal ends **163**, **165** of the tines **162**, **164** penetrate the venous wall **202**B.

(42) In the method of the present disclosure, the target implant site **200** includes a portion of the ventricular tissue of the walls **202**A, **202**B of the coronary sinus where the atrial endocardium can be reached. The coronary sinus is a large vein located in the atrioventricular groove (coronary sulcus) between the left ventricle and the right atrium of the heart. This coronary sinus collects deoxygenated blood from several cardiac veins located around the heart muscle. At its origin, the coronary sinus drains into the right atrium, and ends at its junction with the great cardiac vein. The orifice of the coronary sinus resides at an inferior border of the triangle of Koch (TOK). The apex of the TOK corresponds to the central fibrous body (CFB) of the heart where the His bundle penetrates, and the TOK is bordered posteriorly by a fibrous extension from the Eustachian valve referred to as the tendon of Todaro. The anterior border of the TOK is demarcated by the hinge (annulus) of the septal leaflet of the tricuspid valve. Since the coronary sinus is easily identifiable under sonogram and proximal the TOK, in some patients implanting the pacing capsule **140** within the tissue of the coronary sinus can provide more reproducible implantation while maintaining excellent pacing performance.

(43) The coronary sinus is located in the posterior portion of the coronary sulcus on the diaphragmatic or posterior surface of the heart. The coronary sinus empties directly into the right atrium near the conjunction of the posterior interventricular sulcus and the coronary sulcus (crux cordis area), located between the inferior vena cava and tricuspid valve. This atrial ostium can be partially covered by a Thebesian valve, although the anatomy of this valve may in some cases be variable. The coronary sinus receives drainage from most epicardial ventricular veins, including the oblique vein of the left atrium (and other left and right atrial veins), the great cardiac vein, the posterior vein of the left ventricle, the left marginal vein, and the posterior interventricular vein. For example, the length of the coronary sinus in adults can vary from about 15 mm to about 65 mm.

(44) FIG. **5** is a two-dimensional (2D) ventricular map **300** of a patient's heart (e.g., a top-down view) showing the left ventricle **320** in a standard 16 segment view and the right ventricle **322**. The map **300** includes a plurality of areas **326** corresponding to different regions of a human heart. As illustrated, the areas **326** are numerically labeled 1-16 (e.g., which correspond to 16 segments of the left ventricle of a human heart). Areas **326** of the map **300** may include basal anterior area **1**, basal anteroseptal area **2**, basal inferoseptal area **3**, basal inferior area **4**, basal inferolateral area **5**, basal anterolateral area **6**, mid-anterior area **7**, mid-anteroseptal area **8**, mid-inferoseptal area **9**, mid-inferior area **10**, mid-inferolateral area **11**, mid-anterolateral area **12**, apical anterior area **13**, apical septal area **14**, apical inferior area **15**, and apical lateral area **16**. The inferoseptal and anteroseptal areas of the right ventricle **322** are also illustrated, as well as the right bundle branch (RBB) and left bundle branch (LBB).

(45) Once implanted, leadlets **148** may be positioned in the target implant region **200** (FIG. **4**). With reference to the map **300**, possible implant locations for the distal end **149** of leadlets **148** include the basal inferoseptal area **3**, the basal inferior area **4**, the mid-inferoseptal area **9**, and the mid-inferior area **10**. With reference to map **300**, the septal region includes one or more of the basal anteroseptal area **2**, basal anteroseptal area **3**, mid-anteroseptal area **8**, mid-inferoseptal area **9**, and the apical septal area **14**. In some embodiments, leadlets **148** may be positioned in the area **324** in FIG. **5**.

(46) FIG. **6** is a conceptual diagram of the heart showing the location of the walls **202**A, **202**B of the coronary sinus **200**. The implant location for the pacing capsule **140** can be selected within ventricular tissue along the entire length of the coronary sinus, and the pacing capsule **140** can be

implanted in the shown region of the coronary sinus/great cardiac vein with the distal tip **149** of leadlets **148** penetrating into the basal ventricular myocardium.

(47) Referring now to FIG. **7**, a system **410** includes a catheter delivery system **432** including a catheter body **434** with a distal region **491**. The catheter body **434** can be made of any flexible material, including metals, polymeric materials, and the like. In some examples, the catheter body **434** is formed by extrusion of a polymeric material including, but not limited to, polyethylene (PE), nylon, polypropylene (PP), polyether block amide (PEBA), polybutylene terephthalate (PBT), and combinations thereof. In other examples, the catheter body **434** can be formed by processes including, but not limited to, molding, three-dimensional (3D) printing, additive manufacturing, and the like. In various examples, the catheter body **434** can be formed from a single layer of polymeric material, or multiple layers of the same or different polymeric materials.

(48) In some examples, which are not intended to be limiting, the catheter body **434** has an outside diameter of about 0.100 inches (2.54 mm) to about 0.500 inches (12.7 mm), or about 0.200 inches (5.08 mm). In some examples, the catheter body **434** can optionally include a reinforcing or a catheter deflection material such as, for example, metal strands, ribbons, wires and the like (not shown in FIG. **7**).

(49) A proximal end **431** of the catheter body **434** is connected to a control handle **436** that can be used to deflect the catheter body **434** and deploy a leadless pacing capsule **440**. The pacing capsule **440** is releasably retained in a delivery cup **442** positioned at a distal end **433** of the catheter body **434**. As noted above with reference to FIG. **2**, suitable leadless pacemaker pacing capsules **440** include those available from Medtronic, Inc., Minneapolis, MN, under the trade designation MICRA.

(50) In some examples, the catheter body **434** may be deflected using an optional curve deflection control **452** on the handle **436**. In some examples, during placement procedures the pacing capsule **440** remains tethered via an optional mechanical tether (not shown in FIG. **7**) bound to a tether pin **457** in the control handle **436**.

(51) Once the delivery cup **442** is positioned and temporarily fixed at the proper location within the coronary sinus of the heart, the pacing capsule **440** is deployed from the delivery cup **442** using a deployment control **450** on the handle **436**. The pacing capsule **440** is implanted into the cardiac tissue using a fixation component or implanting mechanism with an arrangement of the memory material metal tines **460**. In the embodiment of FIG. **7**, the tines **460** include a penetrator tine **462** and at least one fixation tine **464**, which are depicted in a retracted position within the delivery cup **442**.

(52) The handle **436** further includes a fluid port **456**, which can provide a fluid flush through the catheter body **434** using a fluid such as, for example, water, saline, and the like.

(53) After the pacing capsule **440** is implanted in the tissue of the heart, a tether lock (not shown in FIG. **7**) on the handle **436** is released and a tether (not shown in FIG. **7**) is removed from a retrieval structure **458** on the pacing capsule **440**, and the catheter body **434** is withdrawn from the vascular system of the patient.

(54) In FIG. **7**, the distal end **433** of the catheter body **434** includes the generally cylindrical and isodiametric delivery cup **442**, which includes a wall **441** configured to securely retain the generally cylindrical pacing capsule **440** of the leadless pacemaker as the catheter body **434** is maneuvered through the vasculature of the patient. In various examples, the delivery cup **442** may be made from a polymeric material that may be the same or different from the polymeric material used to form the tubular body **434**. In some examples, the delivery cup **442** may be made from a high impedance acoustic material, which in the present application refers to materials having an acoustic impedance higher than the acoustic impedance of a target tissue into which the pacing capsule **440** is to be implanted. In some examples, the delivery cup may be made from a low impedance acoustic material, which refers herein to materials having an acoustic impedance lower than the acoustic impedance of the target tissue.

(55) In various examples, the delivery cup **442** may be a portion of the tubular body **434**, may be a separate structure press-fit into the tubular body **434**, or may be a separate structure bonded to the tubular body **434** by an adhesive, ultrasonic welding, overmolding, and the like. Like the catheter body **434**, the delivery cup **442** may be formed by a wide variety of manufacturing processes including, but not limited to, extrusion, molding, 3D printing, additive manufacturing, and the like.
(56) The delivery cup **442** includes a first end **445** that is integral with or connected to the distal end **433** of the tubular body **434** of the catheter **432**. A distal second end **447** of the delivery cup **443** includes an aperture **470** through which the pacing capsule **440** is deployed.
(57) The pacing capsule **440** includes a first proximal end **471** retained in the delivery cup **442** at the first end **445** thereof. A second distal end **447** of the delivery cup **442** is proximal the deployment aperture **470** and includes the metal tines **460**. While retained in the delivery cup **442**, the distal end **473** of the pacing capsule **440**, as well as the tines **460**, are substantially aligned near the distal end **447** of the delivery cup **442**, and the tines **460** do not protrude from the aperture **470**. The wall **441** of the delivery cup **442** maintains the orientation of the tines **460** until the pacing capsule **440** is deployed at a desired pacing capsule implantation site. When the pacing capsule **440** exits the delivery cup **442**, the tines **460**, which are retracted and maintained in position against the wall **441** of the delivery cup, spring out of the aperture **470** and extend laterally from a longitudinal axis of the delivery cup **442**. Once deployed, at least one penetrator tine **462** and at least one fixation tine **464** are configured to pierce a target tissue and maintain the position and orientation of the pacing capsule **440** such that the distal end **449** of the electrode **448** remains embedded in the ventricular tissue of the coronary sinus of the patient.
(58) The delivery cup **442** includes at least one balloon **476**. In FIG. **7**, the balloon **476** is shown in an inflated state, and in some examples, which are not intended to be limiting, has a conical or pear-like shape, but balloons with a wide variety of shapes can be used to fit into a selected region of coronary sinus tissue near a target implant site. The balloon **476** may be attached to an external surface **481** of the catheter **432**, to an external surface **483** of the wall **441** of the delivery cup **442**, or a combination thereof.
(59) In some examples, which are not intended to be limiting, the balloon **476** is formed from a soft, flexible, compliant polymeric material such as, for example, polyethylene (PE)/ethylene vinyl alcohol (EVA) blends, silicone, polyurethane, polyether block amide, thermoplastic elastomer, thermoplastic rubber and combinations thereof. However, in some examples, balloons formed from these materials may be made less compliant, or even relatively non-compliant. A more compliant balloon easily takes on the shape of the coronary sinus of a patient without applying excessive pressure to the vessel (to, for example, avoid dissection). The compliant balloon conforms to the shape necessary to occupy the coronary sinus and push the delivery cup **442** to a desired position to deploy the pacing capsule **440** in a desired orientation within the coronary sinus. In another example, a less compliant balloon with a more specific shape such as, for example, a funnel or a cork, may be more effective in plugging the coronary sinus of a patient during a procedure, may be useful to apply more pressure to the walls of the coronary sinus when the balloon is inflated.
(60) In some embodiments, the balloon **476** is made of an echogenic material, and may be shaped to provide orientation feedback regarding the position of the distal end **447** of the delivery cup **442** on a sonogram image.
(61) The balloon **476** includes a balloon wall **475** that may be formed from a single layer or multiple layers of polymeric materials and may optionally include reinforcing materials to enhance strength and burst resistance. In some examples, the balloon **476** may include at least one outer layer of an echogenic material.
(62) In some examples, which are not intended to be limiting, the balloon **476** has a length of about 1 cm to about 10 cm. The balloon wall **475** can be attached to the external surfaces **481**, **483** by any suitable technique including, for example, bonding, fusing, adhesives, and the like.
(63) As shown schematically in FIG. **8**A, in the embodiment of FIG. **7**, the balloon **476** extends

only a portion of the way around a circumference of the delivery cup **442**. However, in other embodiments, as shown schematically in FIG. **8**B, the balloon **476** may extend around the full circumference of the delivery cup **442**.

(64) In some embodiments (not shown in FIG. **7**), the tubular body **434** or the delivery cup **442** may include multiple balloons **476** thereon, which may have the same shapes and sizes or different shapes and sizes, depending on the configuration of the tissue at an intended desired implant location, the force required to prevent dislodgement of the delivery cup **442** from a region of the coronary sinus during an implantation procedure, and the like.

(65) The deployment of the balloon **476** can also provide a more echogenic shape to the delivery cup **442** without a required increase in the length of the delivery cup **442**. The echogenic balloon is clearly visible and provides less obstructed sonogram images with improved clarity showing both the location of the distal tip **447** of the delivery cup **442** and the cardiac anatomy/tissue of a patient. The unobstructed sonogram images of the balloon provide improved confirmation of the location of the distal end **447** of the delivery cup **442** in the coronary sinus, as well as the implantation status of the pacing capsule **440** in the target cardiac tissue of the patient.

(66) The thin, uninflated balloon **476** minimally increases the diameter of the device **410** during portions of the procedure where minimal diameter is important (for example, to provide improved venous access or to traverse smaller diameter veins).

(67) Referring now to the schematic depiction of FIG. **9**, in operation, after the delivery cup **442** is positioned in a desired implant location **500** within the coronary sinus of the heart, a fluid is introduced into the balloon lumen **493** via, for example, a syringe, and exits a fluid egress port **480** in the balloon wall **475** to inflate and expand the balloon **476**. In various examples, which are not intended to be limiting, the balloon **476** is inflated with an ultrasonically transparent fluid such as water or saline, a non-ultrasonically transparent fluid such as a radio-opaque contrast medium, or a mixture or combination thereof.

(68) Once inflated, the wall **475** of the expanded compliant balloon **476** on a first side of the wall **441**A of the delivery cup **442** conforms to a first wall portion **502**A near a target implant location in the coronary sinus tissue **500**. The expanded balloon wall **475** presses against the first wall portion **502**A along a first direction **490**, which urges a second side of the wall **441**B of the delivery cup **442** along a generally opposed second direction **492** into position against an opposed second wall **502**B at the target implant location. Once the delivery cup **442** is positioned and temporarily fixed in a desired position between the walls **502**A, **502**B at the target implant location, the pacing capsule **440** may be securely deployed from the aperture **470** at the distal end **447** of the delivery cup **442**. Once the pacing capsule **440** emerges from the aperture **470**, the spring-loaded tines **460** spring away and expand laterally from their deflected positions within the delivery cup **442** and move toward the wall **502**B of the coronary sinus.

(69) During and after deployment, the penetrator tine **462** has a deployment stiffness selected such that the tip **463** thereof penetrates the target tissue of the wall **502**B generally along the direction **492**. Following penetration, the tine **462** and the distal end **449** of the electrode **448** embeds in the target tissue within the second wall **502**B. The fixation tine **465** assists in maintaining the position of the pacing capsule **442** against the wall **502**B during and after the implantation procedure.

(70) As shown schematically in FIG. **9**, the tip **463** of the penetrator tine **462**, as well as the tip **465** of the fixation tine **464**, penetrate the target tissue of the coronary sinus wall **502**B along the second direction **492**, which is generally opposed to the first direction **490** along which force is applied against the coronary sinus wall **502**A by the wall **475** of the compliant balloon **476**. The constant force applied by the pressurized balloon **476** against the wall **502**A urges the delivery cup **442** into position against the opposed wall **502**B and thus assists the fixation tine **464** in maintaining the position of the pacing capsule **440** between the opposed walls **502**A, **502**B during the deployment procedure. The force applied by the inflated balloon **476** also resists movement of the delivery cup **442** and the pacing capsule **440** along the direction **490** toward the wall **502**A as the tip **463** of the

penetrator tine **462** pierces and lodges in the target tissue **502**B. The force applied by the compliant balloon **476** can also prevent slippage of the delivery cup **442** or the pacing capsule **440** along the wall **502**B during the deployment procedure. The force applied by the walls **475** of compliant balloon **476** thus enhances the accuracy and reliability of the procedure in which the pacing capsule **440** and the electrode **448** are implanted in the target tissue at the implant location in the coronary sinus tissue **502**B.

(71) Once the pacing capsule **440** is affixed in the implant location in the coronary sinus wall **502**B and released from its tether, the balloon **476** may be at least partially deflated. The delivery cup **442** may then be withdrawn from the vasculature of the patient by removing the affixed catheter **432** (not shown in FIG. **9**).

(72) In another aspect, the present disclosure is directed to a method of implanting an implantable medical device (IMD) such as, for example, a leadless pacemaker, in a coronary sinus of a heart of a patient. As shown in the flow chart of FIG. **10**, the example method **600** includes inserting into an appropriate vein of a patient such as, for example, a femoral vein, a system for delivery of the leadless pacemaker (**602**). The system includes a pacing capsule and a catheter. The catheter includes an elongate flexible tubular body with a distal end having a delivery cup configured to releasably retain the pacing capsule. An outer surface of the delivery cup includes at least one balloon as described in FIGS. **8-9** above. The pacing capsule includes an arrangement of tines configured to pierce and affix the pacing capsule to a target tissue in the coronary sinus of the patient, and embed the distal tip of the electrode in the coronary sinus wall.

(73) While typically performed under fluoroscopy, the example method optionally includes monitoring the location of the delivery cup with an ultrasonic imager to provide a sonogram. Any suitable ultrasonic imaging system may be used, and in some examples the imaging system includes an ultrasonic probe that moves along an external surface of the skin of the patient. In another example, an intracardiac echo (ICE) probe may be inserted into the femoral vein of the patient and maneuvered into position in the right atrium or right ventricle to image the anatomy of the patient. In various examples, which are not intended to be limiting, suitable probe apparatus include ultrasonic probes available from General Electric (GE), Philips, Siemens and the like. In various examples, which are provided by way of example and are not intended to be limiting, the transducers in the ultrasonic probe apparatus operate over a frequency range of about 1 MHz to about 60 MHz, or about 3 MHz to about 10 MHz for imaging procedures, and have a focal length of about 1 cm to about 4 cm, or about 2 cm to about 3 cm.

(74) The example method **600** further includes maneuvering the catheter within the vasculature of the patient to position the delivery cup into an implant location in the coronary sinus of the patient (**604**). In some examples, the location of the delivery cup can be monitored in the sonogram.

(75) The example method **600** further includes inflating the balloon to urge the delivery cup against the target tissue and temporarily anchor the delivery cup at the implant location (**606**).

(76) After the delivery cup is anchored at the implant location by the balloon, the method **600** further includes deploying the pacing capsule from the delivery cup via the catheter **34** (FIG. **1**) or an external tool such that at least some of the tines and the distal tip of the electrode pierce the target tissue and affix the pacing capsule at the implant location (**608**).

(77) The example method **600** further includes at least partially deflating the balloon (**610**) and removing the catheter and the delivery cup from the vasculature of the patient (**612**).

(78) The following examples are illustrative of the techniques described herein.

(79) Example 1: A system for delivery of a leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a proximal end and a distal end, wherein the distal end of the tubular body comprises a delivery cup with an external surface comprising an inflatable compliant balloon; and a pacing capsule of the leadless pacemaker releasably retained in the delivery cup, wherein the pacing capsule comprises an arrangement of tines configured to deploy and pierce a target tissue at a desired pacing capsule implant site in a coronary sinus of a

patient; and wherein the balloon, when at least partially inflated, is configured to urge the delivery cup against the target tissue during deployment of the pacing capsule.

(80) Example 2: The system of claim **1**, wherein the balloon resides on a first side of the delivery cup, and the tines on the pacing capsule are configured to deploy from the delivery cup toward a second side of the delivery cup opposite the first side thereof.

(81) Example 3: The system of claim **1** or **2**, wherein the arrangement of tines comprises at least one penetrator tine with a tip configured to pierce and penetrate the target tissue.

(82) Example 4: The system of claim **3**, wherein the arrangement of tines further comprises at least one fixation tine with a tip configured to pierce and penetrate the target tissue.

(83) Example 5: The system of any one or more of claims **1-4**, wherein the delivery cup comprises a fluid egress port fluidly connected to the balloon.

(84) Example 6: The system of any one or more of claims **1-5**, wherein the balloon comprises a polymeric material chosen from polyethylene (PE), ethylene vinyl alcohol (EVA), silicone, polyurethane, polyether block amide, and mixtures and combinations thereof.

(85) Example 7: The system of any one or more of claims **1-6**, wherein the balloon extends around a portion of a circumference of an external surface of the delivery cup, and wherein the portion of the circumference is less than the entire circumference.

(86) Example 8: The system of any one or more of claims **1-6**, wherein the balloon extends around a circumference of the external surface of the delivery cup.

(87) Example 9: The system of any one or more of claims **1-8**, wherein the delivery cup comprises a plurality of balloons.

(88) Example 10: The system of any one or more of claims **1-9**, wherein the balloon, when inflated with a fluid, has a conical shape.

(89) Example 11: The system of claim **10**, wherein the fluid is chosen from air, water, saline, and mixtures and combinations thereof.

(90) Example 12: The system of any one or more of claims **1-11**, wherein the balloon has a length of about 1 centimeters (cm) to about 10 cm, and an inflated diameter of about 2 cm to about 4 cm.

(91) Example 13: The system of any one or more of claims **1-12**, wherein the tubular body of the catheter comprises a polymeric material chosen from polyethylene (PE), nylon, silicone, polyurethane, polyether block amide, and combinations thereof.

(92) Example 14: The system of any one or more of claims **1-13**, wherein the tubular body comprises a metal reinforcing material.

(93) Example 15: The system of any one or more of claims **1-14**, wherein the tubular body further comprises a braid, a coil, or combinations thereof.

(94) Example 16: The system of any one or more of claims **1-15**, wherein the tubular body further comprises a pull wire.

(95) Example 17: A method for implanting a leadless pacemaker in a target cardiac tissue, the method comprising: inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup comprises an inflatable compliant balloon; and a pacing capsule releasably retained in the delivery cup, wherein the pacing capsule comprises an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue, wherein the target cardiac tissue is a wall of a coronary sinus; maneuvering the delivery cup in the vasculature of the patient to an implant location in the coronary sinus of the patient; at least partially inflating the balloon to urge the delivery cup against the target cardiac tissue and anchor the delivery cup at the implant location; and deploying the pacing capsule from the delivery cup such that the tines pierce the target tissue to embed the electrode in the target tissue and affix the pacing capsule in the target tissue at the implant location.

(96) Example 18: The method of claim **17**, further comprising: deflating the balloon; and removing the catheter and the delivery cup from the vasculature of the patient.

(97) Example 19: The method of claim **17** or **18**, further comprising: monitoring the location of the delivery cup with an ultrasonic imager to form a sonogram image; and maneuvering the delivery cup in the vasculature of the patient as shown in the sonogram image into the coronary sinus.
(98) Example 20: The method of any one or more of claims **17-19**, wherein the balloon overlies at least a portion of the external surface of the delivery cup.
(99) Example 21: The method of any one or more of claim **17-20**, wherein the balloon extends around a circumference of the external surface of the delivery cup.
(100) Example 22: The method of claim **21**, wherein the balloon extends around a portion of a circumference of the external surface of the delivery cup, and wherein the portion of the circumference is less than the entire circumference.
(101) Example 23: The method of any one or more of claim **17-22**, wherein the distal end of the catheter comprises a plurality of balloons.
(102) Example 24: The method of claim any one or more of **17-23**, wherein the balloon, when inflated with a fluid, has a conical shape.
(103) Example 25: The method of any one or more of claims **17-24**, wherein the balloon, when inflated, applies a force against a first wall of the coronary sinus, and the tines on the pacing capsule, deploy laterally from the delivery cup into a second wall of the coronary sinus opposite the first wall thereof.
(104) Example 26: The method of claim **25**, wherein the balloon urges the delivery cup against the second wall of the coronary sinus.
(105) Example 27: The method of any one or more of claims **17-26**, wherein the balloon is sufficiently compliant to avoid damage to a wall of the coronary sinus.
(106) Example 28: A method for implanting a leadless pacemaker in a target cardiac tissue in a coronary sinus of a patient, the method comprising: inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup comprises an inflatable balloon; and a pacing capsule releasably retained in the delivery cup, wherein the pacing capsule comprises an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue; maneuvering the delivery cup in the vasculature of the patient to an implant location in a first wall of the coronary sinus of the patient; inflating the balloon to urge the delivery cup against a second wall of the coronary sinus opposite a first wall and temporarily affix the delivery cup at the implant location; and deploying the pacing capsule from the delivery cup such that the tines pierce and embed in the first wall to implant the pacing capsule in the first wall, and the electrode embeds in the first wall.
(107) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A system for delivery of a leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a proximal end and a distal end, wherein the distal end of the tubular body comprises a delivery cup with an external surface comprising an expandable member; and a pacing capsule of the leadless pacemaker releasably retained in the delivery cup, wherein the pacing capsule comprises an arrangement of tines comprising memory material configured to deploy and pierce a target tissue at a desired pacing capsule implant site in a coronary sinus of a patient; and wherein the expandable member, when at least partially expanded, is configured to apply a force against a first wall of the coronary sinus to urge the delivery cup against the target tissue during deployment of the pacing capsule, and the tines on the pacing capsule are configured to deploy laterally from the delivery cup into a second wall of the coronary sinus opposite the first wall thereof when the delivery cup is urged against the target tissue.

2. The system of claim 1, wherein the expandable member resides on a first side of the delivery cup, and the tines on the pacing capsule are configured to deploy from the delivery cup toward a second side of the delivery cup opposite the first side thereof.
3. The system of claim 1, wherein the arrangement of tines comprises at least one penetrator tine with a tip configured to pierce and penetrate the target tissue.
4. The system of claim 3, wherein the arrangement of tines further comprises at least one fixation tine with a tip configured to pierce and penetrate the target tissue.
5. The system of claim 1, wherein the delivery cup comprises a fluid egress port fluidly connected to the expandable member.
6. The system of claim 1, wherein the expandable member comprises a balloon that comprises a polymeric material chosen from polyethylene (PE), ethylene vinyl alcohol (EVA), silicone, polyurethane, polyether block amide, and mixtures and combinations thereof.
7. The system of claim 1, wherein the expandable member extends around a portion of a circumference of an external surface of the delivery cup, and wherein the portion of the circumference is less than the entire circumference.
8. The system of claim 1, wherein the expandable member extends around a circumference of the external surface of the delivery cup.
9. The system of claim 1, wherein the delivery cup comprises a plurality of expandable members.
10. The system of claim 1, wherein the expandable member, when expanded, has a conical shape.
11. The system of claim 1, wherein the expandable member has a length of about 1 centimeters (cm) to about 10 cm, and an expanded diameter of about 2 cm to about 4 cm.
12. The system of claim 1, wherein the tubular body of the catheter comprises a polymeric material chosen from polyethylene (PE), nylon, silicone, polyurethane, polyether block amide, and combinations thereof.
13. The system of claim 1, wherein the tubular body comprises a metal reinforcing material.
14. The system of claim 1, wherein the tubular body further comprises a braid, a coil, or combinations thereof.
15. The system of claim 1, wherein the tubular body further comprises a pull wire.
16. A method for implanting a leadless pacemaker in a target cardiac tissue, the method comprising: inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup comprises an expandable member; and a pacing capsule releasably retained in the delivery cup, wherein the pacing capsule comprises an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue, wherein the target cardiac tissue is a wall of a coronary sinus; maneuvering the delivery cup in the vasculature of the patient to an implant location in the coronary sinus of the patient; at least partially expanding the expandable member to urge the delivery cup against the target cardiac tissue and anchor the delivery cup at the implant location, wherein the expandable member, when expanded, applies a force against a first wall of the coronary sinus; and deploying the pacing capsule from the delivery cup so that the tines deploy laterally from the delivery cup and pierce into the target tissue to embed the electrode in the target tissue and affix the pacing capsule in the target tissue at the implant location, wherein the target tissue is a second wall of the coronary sinus opposite the first wall thereof.
17. The method of claim 16, further comprising: contracting the expandable member; and removing the catheter and the delivery cup from the vasculature of the patient.
18. The method of claim 16, further comprising: monitoring the location of the delivery cup with an ultrasonic imager to form a sonogram image; and maneuvering the delivery cup in the vasculature of the patient into the coronary sinus based on the sonogram image.
19. The method of claim 16, wherein the expandable member overlies at least a portion of the external surface of the delivery cup.

20. The method of claim 16, wherein the expandable member extends around a circumference of the external surface of the delivery cup.
21. The method of claim 20, wherein the expandable member extends around a portion of a circumference of the external surface of the delivery cup, and wherein the portion of the circumference is less than the entire circumference.
22. The method of claim 16, wherein the distal end of the catheter comprises a plurality of expandable members.
23. The method of claim 16, wherein the expandable member, when expanded, has a conical shape.
24. The method of claim 16, wherein the expandable member urges the delivery cup against the second wall of the coronary sinus.
25. The method of claim 16, wherein the expandable member is sufficiently compliant to avoid damage to a wall of the coronary sinus.
26. The method of claim 16, wherein the expandable member comprises an inflatable balloon.
27. A method for implanting a leadless pacemaker in a target cardiac tissue in a coronary sinus of a patient, the method comprising: inserting into a vasculature of a patient a system for delivery of the leadless pacemaker, the system comprising: a catheter comprising an elongate flexible tubular body with a distal end having a delivery cup, and wherein an external surface of the delivery cup comprises an expandable member; and a pacing capsule releasably retained in the delivery cup, wherein the pacing capsule comprises an electrode and an arrangement of tines configured to affix the pacing capsule to the target cardiac tissue; maneuvering the delivery cup in the vasculature of the patient to an implant location in a first wall of the coronary sinus of the patient; expanding the expandable member to urge the delivery cup against the first wall of the coronary sinus opposite a second wall and temporarily affix the delivery cup at the implant location, wherein the expandable member, when expanded, applies a force against the second wall of the coronary sinus; and deploying the pacing capsule from the delivery cup so that the tines deploy laterally from the delivery cup into the first wall to pierce and embed in the first wall to implant the pacing capsule in the first wall, and the electrode embeds in the first wall.